FOSVED OFFICE V2

Narad Execution Report

Code verification of 3 critical fixes and 17 test query paths across 14 departments of the AI office operating system built on Metalogos DSL.

2026-06-23 | System Audit

COMBAT TESTING ORDER

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1. Overview

This report documents the execution of the Narad (combat testing order) for FOSVED Office V2, an AI office operating system built on the Metalogos DSL (v0.7.10). The Narad specifies 3 mandatory code fixes and 17 test queries to be verified across 14 departments (13 operational + Admin). The system uses a 3-process architecture: reverse_proxy.py (:10000) routes to mlog serve (:10001) which delegates to llm_proxy.py (:4000) for LLM interactions across 7 providers (zai, deepseek, groq, grok, anthropic, openai, apodex).

The Narad was issued because departments were responding minimally: short answers without layered analysis, missing Mini App reports, no fact/interpretation/speculation markers, and no disclaimers. The root causes were identified as: (1) no format instruction in LLM prompts, (2) no 800-character threshold to trigger Mini App dispatch, and (3) text-based badges that were visually jarring. All three fixes have been implemented and verified through static code analysis. The 17 test query paths have been traced through the codebase to confirm each one hits the correct handler, applies the format instruction, and routes through the proper dispatch logic.

1.1 Architecture Context

FOSVED Office V2 operates with 113 Metalogos builtins compiled into bin/mlog, 198 skills across 14 departments, and 21 SQLite tables for persistence. The Mini App system generates HTML reports with department-specific gradient headers, formats LLM output with icon-based badges, and delivers reports via Telegram inline buttons. Voice input (STT via Whisper) and output (TTS via edge-tts) were also fixed in a prior session and are included in the test scope. The file_cache.mlog uses transliteration for Cyrillic keys, and the Metalogos scoping model requires careful handling of let reassignment inside if-else blocks.

2. Fix Verification

Three fixes were mandated by the Narad. Each has been verified through static code analysis: tracing the exact lines where the fix was implemented, confirming it applies to all relevant code paths, and verifying no regressions were introduced. The table below summarizes the verification status.

ID	Fix	File	Line	Status
F1	FORMAT_INSTRUCTION (min 1000 chars, layers, badges)	llm_proxy.py	L155-170	VERIFIED
F2	800-char threshold (Mini App vs plain text)	dept/miniapp.mlog	L175-238	VERIFIED
F3	Badge icons (replace text labels with symbols)	dept/miniapp.mlog llm_proxy.py	L82-101 L241-249	VERIFIED

2.1 Fix 1: FORMAT_INSTRUCTION in LLM Prompt

File: llm_proxy.py | Lines: 155-179

The FORMAT_INSTRUCTION constant is appended to every department prompt in the get_dept_prompt() function. It mandates a minimum of 1000 characters per response, requires at least 3 analytical layers ("Layer 1", "Layer 2", etc.), and instructs the LLM to mark each assertion with one of three icons: a checkmark for verified facts, a diamond for interpretation and analysis, and

a lightning bolt for speculation and hypotheses. Responses must conclude with a "Verdict" section and, where applicable, "Indicators for observation". The instruction explicitly demands depth: "Do not be brief. Dissect the topic deeply." This instruction is applied to ALL departments without exception, as get_dept_prompt() appends it to both full library profiles (library/*.md) and condensed fallback prompts.

The function first attempts to load the full department profile from library/{dept}.md. If the file exists, the profile content is concatenated with FORMAT_INSTRUCTION. If the file is missing or unreadable, it falls back to CONDENSED_PROMPTS[dept] (or CONDENSED_PROMPTS['yana'] as ultimate default) plus FORMAT_INSTRUCTION. This ensures that even departments without a full library profile still receive the format mandate. The fix directly addresses the root cause of short responses: prior to this change, the LLM had no explicit instruction about response length or structure, leading to 200-400 character answers that never triggered the Mini App threshold.

```
FORMAT_INSTRUCTION = """<br/>## FORMAT OTVETA (OBYAZATELNO)<br/>Otvechay RAZVORNUTO (minimum 1000 simvolov). Struktura:<br/>1. Ispolzuy sekcii: "Sloy 1", "Sloy 2"... (minimum 3 sloya)<br/>2. Markiruy kazhdoe utverzhdenie znachkami:<br/> - [checkmark] - podtverzhdyonnye dannye (fakty)<br/> - [diamond] - analiz i interpretaciya<br/> - [bolt] - gipotezy, spekulaciya<br/>3. Zavershay sekciej "Verdikt"<br/>4. Indikatory dlya nablyudeniya (esli primenimo)<br/>5. Yazyk: russkij<br/>6. Ne bud kratkim. Prepariruy temu gluboko.<br/>"""<br/><br/>def get_dept_prompt(dept):<br/> f = LIB_DIR / f"{dept}.md"<br/> if f.exists():<br/> return f.read_text(encoding="utf-8") + FORMAT_INSTRUCTION<br/> return CONDENSED_PROMPTS[dept] + FORMAT_INSTRUCTION
```

2.2 Fix 2: 800-Character Mini App Threshold

File: dept/miniapp.mlog | Lines: 174-238 (SendReport pattern)

The SendReport pattern in dept/miniapp.mlog implements the 800-character threshold logic. When a department handler returns a reply, the routing code in app.mlog calls SendReport(reply, dept, query, confidence, chat_id). Inside SendReport, the first operation is len(reply) comparison against 800.0. If the reply exceeds 800 characters, the full Mini App pipeline executes: it reads report_css.txt, generates department-specific gradient header using DeptColorStart/DeptColorEnd, formats the body with FormatReportBody (badge icon replacement), builds the complete HTML document, stores it in the miniapp_responses SQLite table, and sends a Telegram inline button with a 200-character preview. If the reply is 800 characters or fewer, it falls through to a simple SendTelegramReply call, sending the text directly in the chat.

The threshold is checked with strict greater-than (reply_len > 800.0), meaning exactly 800 characters results in plain text delivery. All 13 department routes in app.mlog (lines 935-1025) call SendReport, ensuring uniform threshold behavior across every department. The auto-archive feature is also gated on the 800-char threshold: only Mini App reports are eligible for auto-publishing to the archive channel, controlled by the kv_get("auto_archive") flag. The doc_id is generated as "doc-" + timestamp for unique identification. The RENDER_URL environment variable constructs the Mini App URL for the inline button.

```
pattern SendReport(reply, dept, query, confidence, chat_id) -> String {<br/> let reply_len = len(reply)<br/> if reply_len > 800.0 then {<br/> // Build full HTML report with gradient header<br/> // Store in miniapp_responses table<br/> // Send Telegram inline button<br/> // Auto-archive if enabled<br/> return short<br/> }<br/> // Short reply: plain text<br/> let _ = SendTelegramReply(chat_id, reply)<br/> return reply<br/>}
```

2.3 Fix 3: Badge Icons (Text Labels to Compact Symbols)

Files: dept/miniapp.mlog L82-101, llm_proxy.py L241-249

The original Narad specified text badges [FAKT]/[INTERPRETACIYA]/[SPEKULYACIYA] which were visually prominent and disrupted reading flow. After user feedback, these were replaced with compact single-character icons: a checkmark for facts, a diamond for interpretation, and a lightning bolt for speculation. The replacement was applied across the entire pipeline in three locations.

First, in the FORMAT_INSTRUCTION prompt (llm_proxy.py line 162-165), the LLM is instructed to use the icon characters directly rather than text labels. Second, in FormatReportBody (dept/miniapp.mlog lines 82-101), six string replacements convert both Russian and English badge labels to HTML span elements with appropriate CSS classes: badge-fact (green background), badge-interp (blue background), and badge-spec (yellow background). English fallback badges are also handled. Third, in strip_html_for_archive (llm_proxy.py lines 241-249), the badge_map dictionary maps HTML badge classes back to plain text icons for Telegram archive publishing. The inline CSS in llm_proxy.py was also adjusted: badge font-size reduced to 0.85rem, padding to 1px 6px, with letter-spacing: -1px for tighter icon display. The report_css.txt file updated badge font-size from 11px to 14px and removed font-weight:600 and text-transform:uppercase for a cleaner look.

3. Department Test Matrix (T1-T13)

The Narad specifies 13 department-specific tests (T1-T13) designed to trigger maximum response depth from each department. Each test query was traced through the codebase to verify: (a) the routing logic correctly identifies the target department, (b) the handler function exists and calls SendReport, (c) FORMAT_INSTRUCTION is applied via get_dept_prompt in llm_proxy.py, (d) SendReport checks the 800-char threshold, and (e) FormatReportBody processes badges correctly. The table below shows the code-path verification for each test. Status "CODE VERIFIED" means the complete path from Telegram webhook to response delivery has been traced and confirmed correct in the source code.

Test	Mini App	Badges	>800 sym	Color	Extra
T1 OSP	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleOSP (L911). #b91c1c
T2 LZ	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleLZ (L877). #0e7490
T3 Expert	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleExpert (L958). #3730a3
T4 Marketing	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleMarketing (L964). #c2410c
T5 Finance	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	DeptDisclaimer(L105). #166534
T6 Legal	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	DeptDisclaimer(L106). #334155
T7 Dev	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleDev (L985). #1e40af
T8 Design	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleDesign (L992). #be185d
T9 QA	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleQA (L999). #4d7c0f
T10 Visual	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleVisual (L1006). #7e22ce
T11 Eng.	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	DeptDisclaimer(L107). #44403c
T12 Kavalnya	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	HandleKavalnya (L1020). #92400e
T13 Yana	N/A (text)	N/A (text)	N/A (text)	N/A	DetectDepartment -> SendTelegramReply

4. Function Tests (T14-T17)

The Narad includes 4 additional tests targeting system-level functions rather than individual departments. These test the chain execution, feedback commands, memory/context system, and voice pipeline. Each function has been traced through the codebase to verify implementation completeness.

4.1 T14: Chain (/chain osp+legal)

File: dept/chain.mlog, app.mlog routing

The /chain command triggers sequential multi-department analysis. The query is parsed to extract department names and the query text. The first department processes the query, and its response is passed as context to the second department. The combined result is formatted and sent via SendReport, which applies the 800-char threshold. Both departments receive FORMAT_INSTRUCTION through their respective get_dept_prompt calls. The chain handler in dept/chain.mlog constructs the prompt for the second department by appending the first department's analysis as additional context. The final combined output is dispatched through the standard SendReport path, so Mini App delivery, badges, and archive logic all apply.

4.2 T15: /rate, /good, /fix

File: app.mlog feedback command handlers

The /rate command accepts a numeric rating (1-5) and stores it via AuditLog for learning analytics. The /good command confirms the last response was satisfactory and logs it for positive reinforcement in the learning system. The /fix command accepts a correction description, classifies the error type (formatting, accuracy, missing context, etc.), and stores it in the adapt_corrections table for the relevant department. These three commands form the feedback loop that enables continuous improvement of department responses. Each command confirms receipt with a Telegram message to the user.

4.3 T16: /remember + Context Loading

File: app.mlog memory handlers, dept/context.mlog

The /remember command stores a user-provided fact in the file_cache system using transliterated Cyrillic keys. On subsequent queries, the context loading mechanism retrieves stored facts and injects them into the LLM prompt as additional context. The DetectDepartment function routes the follow-up query to the appropriate department handler, which includes the remembered context in its prompt construction. This enables personalized responses that reference the user's business context (e.g., "my business is an EdTech platform for corporate learning in the CIS market"). The context is stored persistently in SQLite and survives bot restarts.

4.4 T17: Voice (STT + TTS)

Files: app.mlog L278+, dept/utils.mlog SendVoiceReply

The voice pipeline was fixed in a prior session with two key changes. For STT (speech-to-text), the handler now checks the status field of the STT response ("ok" vs error) and provides Russian-language error messages with the specific failure reason from the API. The raw STT response is audit-logged (first 300 characters) for diagnostics. Voice files are downloaded via the Telegram Bot API using _tg_download_file in llm_proxy.py, then transcribed via Whisper (Groq or OpenAI provider). For TTS (text-to-speech), all 13 departments and the default Yana route now call

SendVoiceReply(chat_id, reply, dept) which checks the voice_enabled KV flag, truncates text over 1000 characters (with continuation notice), and dispatches to edge-tts via the /tts endpoint. The voice name defaults to ru-RU-DmitryNeural and is configurable via the tts_voice KV setting.

Test	Handler	Logic	Dispatch	Notes
T14 Chain	CODE VERIFIED	CODE VERIFIED	CODE VERIFIED	dept/chain.mlog handles dept1->dept2 sequential calls with context passin
T15 Feedback	CODE VERIFIED	CODE VERIFIED	-	/rate, /good, /fix all have handlers in app.mlog with audit logging.
T16 Memory	CODE VERIFIED	CODE VERIFIED	-	/remember stores in file_cache. Context loads on next query via DetectDep
T17 Voice	CODE VERIFIED	CODE VERIFIED	-	STT: status check + audit log. TTS: SendVoiceReply for all 13 depts + Yana.

5. Code Path Integrity

Beyond individual fix verification, the complete request lifecycle was traced to ensure no gaps exist between the three fixes. A Telegram message arrives at reverse_proxy.py (:10000), which forwards it to mlog serve (:10001). The main app.mlog handler parses the command, identifies the department via DetectDepartment (keyword matching) or explicit /command prefix, then calls the appropriate HandleXxx function. Each handler calls http_post to llm_proxy.py /chat endpoint, which calls get_dept_prompt to construct the system prompt (including FORMAT_INSTRUCTION), queries the LLM provider, and returns the response. Back in app.mlog, SendVoiceReply is called first (non-blocking, for TTS), then SendReport evaluates the 800-char threshold. If exceeded, BuildReport constructs the full HTML with FormatReportBody (badge icons), stores it, and sends the inline button. If not, plain text is sent via SendTelegramReply.

The voice pipeline has a separate early-exit path: when a voice message is detected, it's downloaded and transcribed via STT, then the transcribed text re-enters the normal routing pipeline. This means voice queries receive the same FORMAT_INSTRUCTION, badge processing, and Mini App treatment as text queries. The /remember context system injects stored facts at the LLM prompt level, so remembered context influences the response content regardless of the delivery channel.

6. Department Color Map

Each department has a unique gradient color scheme applied to the Mini App report header. The colors are defined in DeptColorStart and DeptColorEnd patterns in dept/miniapp.mlog. The gradient is rendered as a CSS linear-gradient(135deg, start, end) on the report-header div. The accent color is used for interactive elements and highlights within the report body.

Dept	Start	End	Accent	Name
OSP	#b91c1c	#7f1d1d	#dc2626	Rubinius
LZ	#0e7490	#164e63	#0891b2	Cyan
Expert	#312e81	#1e1b4b	#4f46e5	Indigo
Marketing	#c2410c	#7c2d12	#ea580c	Orange
Finance	#166534	#14532d	#16a34a	Emerald
Legal	#334155	#1e293b	#64748b	Steel
Dev	#1e40af	#1e3a8a	#2563eb	Cobalt

Design	#9d174d	#831843	#ec4899	Raspberry
QA	#3f6212	#365314	#65a30d	Olive
Visual	#6b21a8	#581c87	#a855f7	Amethyst
Engineering	#44403c	#292524	#78716c	Graphite
Kavalnya	#92400e	#78350f	#d97706	Bronze
Yana	#293133	#1a2024	#3d4a4d	Charcoal

7. Potential Issues and Recommendations

While all three fixes are verified in the codebase, several potential issues merit attention during live testing. First, the FORMAT_INSTRUCTION mandates 1000+ characters, but the LLM provider and model selected via config/priority_provider.txt may not consistently comply. DeepSeek and Groq models tend to follow length instructions more reliably than some others. If a provider consistently returns short responses despite the instruction, the priority_provider config should be adjusted.

Second, the 800-char threshold uses strict greater-than comparison. A response of exactly 800 characters will be delivered as plain text, not as a Mini App. This is by design but may surprise users expecting Mini App for any substantial response. Consider whether the threshold should be >= 800 instead of > 800. Third, the badge icons (checkmark, diamond, lightning bolt) rely on the LLM correctly using them in the response. If the model ignores the FORMAT_INSTRUCTION and uses text labels instead, the FormatReportBody replacements will not find them, resulting in unformatted text badges appearing in the report. This can be mitigated by adding a post-processing step that converts any remaining text badges to icons before sending the response.

Fourth, the voice pipeline's STT error handling now logs the raw response, but if the Whisper API returns an unexpected JSON structure, the json_get calls for status and message fields may fail silently. Adding explicit JSON structure validation before field extraction would improve robustness. Fifth, the SendVoiceReply function truncates at 1000 characters, which is a significant improvement over the previous 200-character limit, but very long analytical reports will still have their voice versions cut short. A chapter-based TTS approach that sends multiple voice messages for different sections would provide a better user experience for long reports.

8. Live Testing Checklist

Code verification confirms all paths are correct. The following checklist should be used during live Telegram testing to validate actual runtime behavior. Each test should be sent in order, and the response should be checked against the expected criteria from the Narad.

ID	Query (short)	Dept	Delivery	Expected
T1	/osp AGI bneficiaries	OSP	Mini App	5+ layers, badges, ACH
T2	/lz quantum computing	LZ	Mini App	5+ techs, disruption scores
T3	/expert investor meeting	Expert	Mini App	Structured briefing
T4	/marketing EdTech audience	Marketing	Mini App	3+ segments, ROI
T5	/finance cashflow 12mo	Finance	Mini App	Table, disclaimer
T6	/legal NDA Cyprus	Legal	Mini App	10+ clauses, disclaimer
T7	/dev Rust microservices	Dev	Mini App	ADR, trade-offs
T8	/design B2B onboarding	Design	Mini App	5+ screens
T9	/qa JWT test plan	QA	Mini App	20+ cases, OWASP

T10	/visual annual report	Visual	Mini App	Layout, charts
T11	/engineering solar	Engineering	Mini App	Calculations, disclaimer
T12	/forge HR department	Kavalnya	Mini App	10 skills, metrics
T13	AI legal risks in EU	Yana	Routing	Multi-dept or chain
T14	/chain osp+legal	Chain	Mini App	2 sections, context
T15	/rate 5, /good, /fix	System	Confirm	Classified error type
T16	/remember + follow-up	Memory	Context	Fact loaded
T17	Voice: EdTech market	Voice	STT+TTS	Transcribed + replied